

CADSmith

Describe a part in plain language. Get CAD geometry a workshop can use.

Presenter: [name, role, organisation] **Event:** [event, date, stand] **Languages:** English and Japanese

OVERVIEW

CADSmith turns a natural-language description of a mechanical part into real CAD geometry. Five AI agents plan the part, write CadQuery code and build it in the OpenCASCADE kernel; the kernel then measures what was built and the part is rejected and rebuilt if the dimensions do not match what was asked for. The difference from ordinary text-to-CAD is that nothing is taken on trust: a plate that should carry four holes and carries one is blocked by measurement, however confident the model was. Parts covered by a published standard — bolts, bearings, gears, shafts — are served from a catalogue instead, exactly and instantly, because a model can only introduce error into a specified figure.

WHAT A VISITOR SEES

- **A live application, not a video.** A part appears in 3D in about two seconds, with its CadQuery source beside it, and its bounding box, volume and watertightness measured by the kernel that built it.
- **A part that stays editable.** Every named value in the script becomes a slider; letting go rebuilds the part in about a second.
- **Something to take away.** A dimensioned A3 sheet with four views and a title block, plus STEP, STL, DXF and GLTF downloads.
- **The whole thing in Japanese.** Interface, run log, verdicts and drawing title block — and Japanese edits and catalogue requests need no model call.

KEY MESSAGES

- Generation is no longer the hard part. Verification is — and verification has to be measurement, not a second opinion from another model.
- Some parts should never be generated at all. Eighteen families of standard part are served from published dimensions, with no model call and no tokens spent.
- The output is parametric source code, so the visitor's own change is one drag away rather than a new run.
- Failures stay on screen: every attempt is kept with the reason it was rejected, which is what makes the successes worth believing.

EVIDENCE

- 32 prompts run end to end against a small open model (Qwen3-VL-8B) on one laptop: of the 25 that reached the model, 22 produced a valid solid and 14 were correct on every measured check; 95 of 112 dimensional checks passed. Catalogue parts: 7 of 7, mean 2.2 s, zero tokens.
- The underlying research pipeline, published by its authors, reaches 100% executable output and cuts mean Chamfer Distance from 28.37 mm to 0.74 mm against a single-call baseline on a 100-prompt benchmark.

WHO IT IS FOR

Mechanical and manufacturing engineers, CAD and PLM teams, design-automation and R&D groups, and anyone weighing up generative AI for physical products. No CAD skill is needed to try it — visitors are invited to type a part in their own words.

FORMAT AND REQUIREMENTS

Duration	60 seconds for the core demo; 5 minutes for the full walkthrough; continuous throughout the day.
Staffing	One presenter at the stand.
Hardware	One laptop (Python 3.11, no GPU) and a table deep enough for a visitor to type on. An external monitor or stand display is welcome; a mouse is needed for the 3D view.
Network	Outbound HTTPS for the model backend. If the venue network fails, the catalogue, the sliders, the drawings, the exports and recorded run replays all work with no network and no API key.

ATTRIBUTION

The CADSmith pipeline and its benchmark are the work of Jesse Barkley, Rumi Loghmani and Amir Barati Farimani, Carnegie Mellon University (arXiv:2603.26512), and are used unmodified. **What is exhibited here is the application built around that pipeline:** the interface, the standard-parts catalogue, the measured specification checks, the engineering drawings, the parametric editing, the spend ceiling and the bilingual layer.

Contact: [email] · [repository or QR destination]